AI5707 – Pattern Recognition and Machine Learning

Assignment 6: Generation of data using 2D Gaussian mixture model, K-Means clustering, Classification

Due: Oct 16 2026

1 Objective

In this assignment, you will learn to generate data corresponding to Gaussian mixture model for two classes, where the density function is given by:

$$p(\mathbf{x}) = \Sigma_{i=1}^{K} \pi_i \mathcal{N}(\mathbf{x}; \mu_\mathbf{i}, \mathbf{C}_i)$$

and perform K-means clustering.

2 Problem definition

A density function as defined in Section 1 is to be generated as indicated below:

1. The values $K, \pi_i, \mu_\mathbf{i}, \mathbf{C}_i$ are chosen empirically. Assume that $\mathbf{x} = [x_1, x_2]^t$ is a 2D vector.
2. Experiment with different $\mathbf{C}_i$, that is, $\mathbf{C}_i = \sigma^2\mathbf{I}$, $\mathbf{C}_{ii}$ is a diagonal covariance matrix and $\mathbf{C}_i$ is a full convariance matrix.
3. Plot data points belonging to different clusters using different colours.
4. Generate data for two different classes, where each class is represented by a mixture density function.
5. Divide the data into 80-20 ratio. Perform K-means clustering on 80% of the data for each class. Plot the boundaries between the clusters of each class.
6. Next, classify using the K-means clusters on the remaining 20% using distortion as a measure.
7. Report your observations.